Pipelined RISC-V Processor

E154 System-On-Chip
Max Conine and Pierce Clark
Apr 30, 2026



Microarchitecture Specification

Our design features a 5-stage processor microarchitecture

RV32IM instruction set

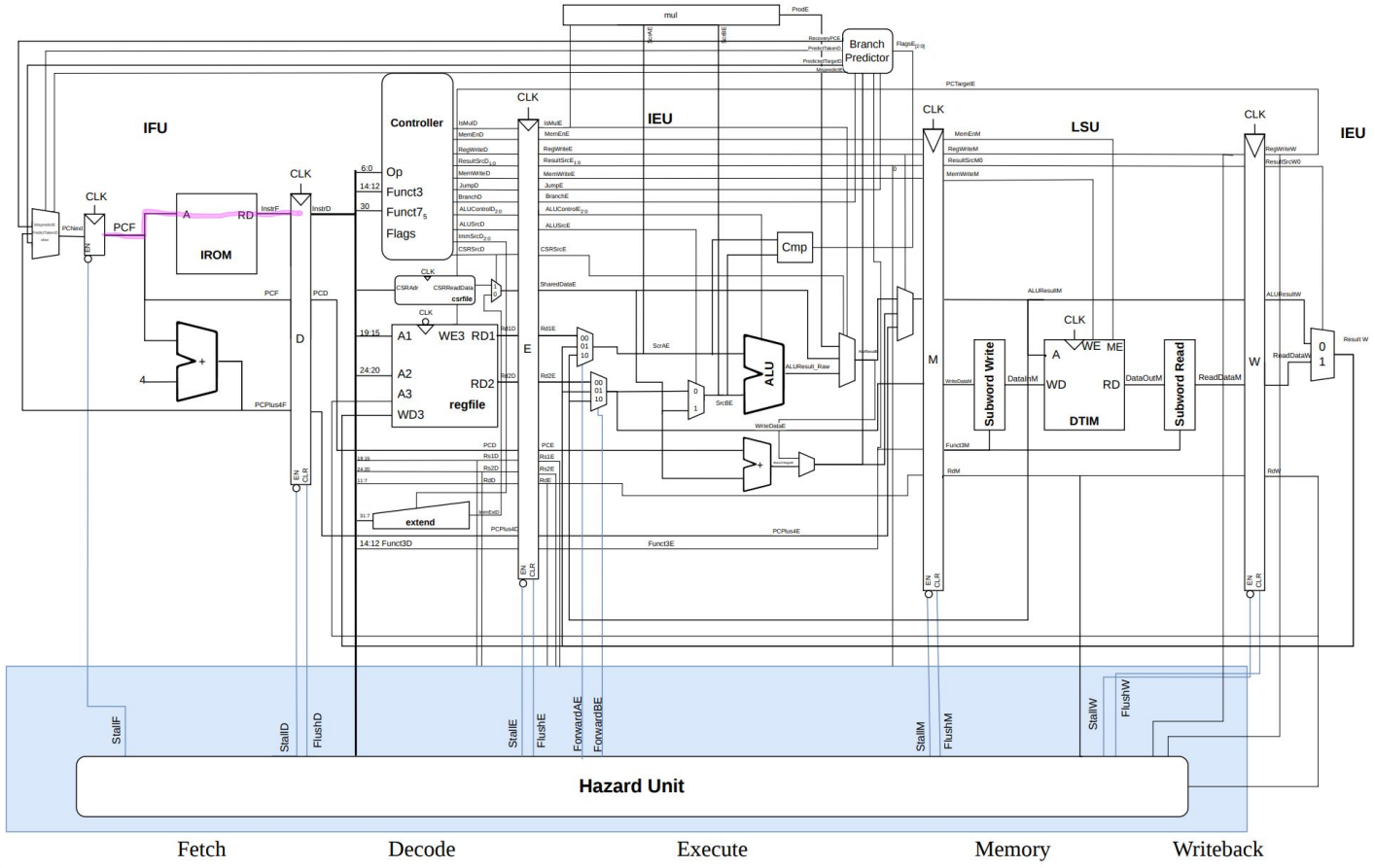
Dedicated 3-stage multiplication unit

Comprehensive hazard unit logic

Backwards Taken Branch Predictor

Inspired by the pipelined processor presented in “Digital Design and Computer Architecture, RISC-V Edition” by David and Sarah Harris as well as the design of the Wally RISC-V Processor.

Top-level block diagram



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Multiply Unit Logic

Let A' = sum(i=0, XLEN-2, A[i]*2^i)

Unsigned: A = A' + A[XLEN-1]*2^(XLEN-1)

Signed: A = A' - A[XLEN-1]*2^(XLEN-1)

Multiplication: A*B

Let P' = A' * B'

    PA = (A' * B[XLEN-1])

    PB = (B' * A[XLEN-1])

    PP = A[XLEN-1] * B[XLEN-1]

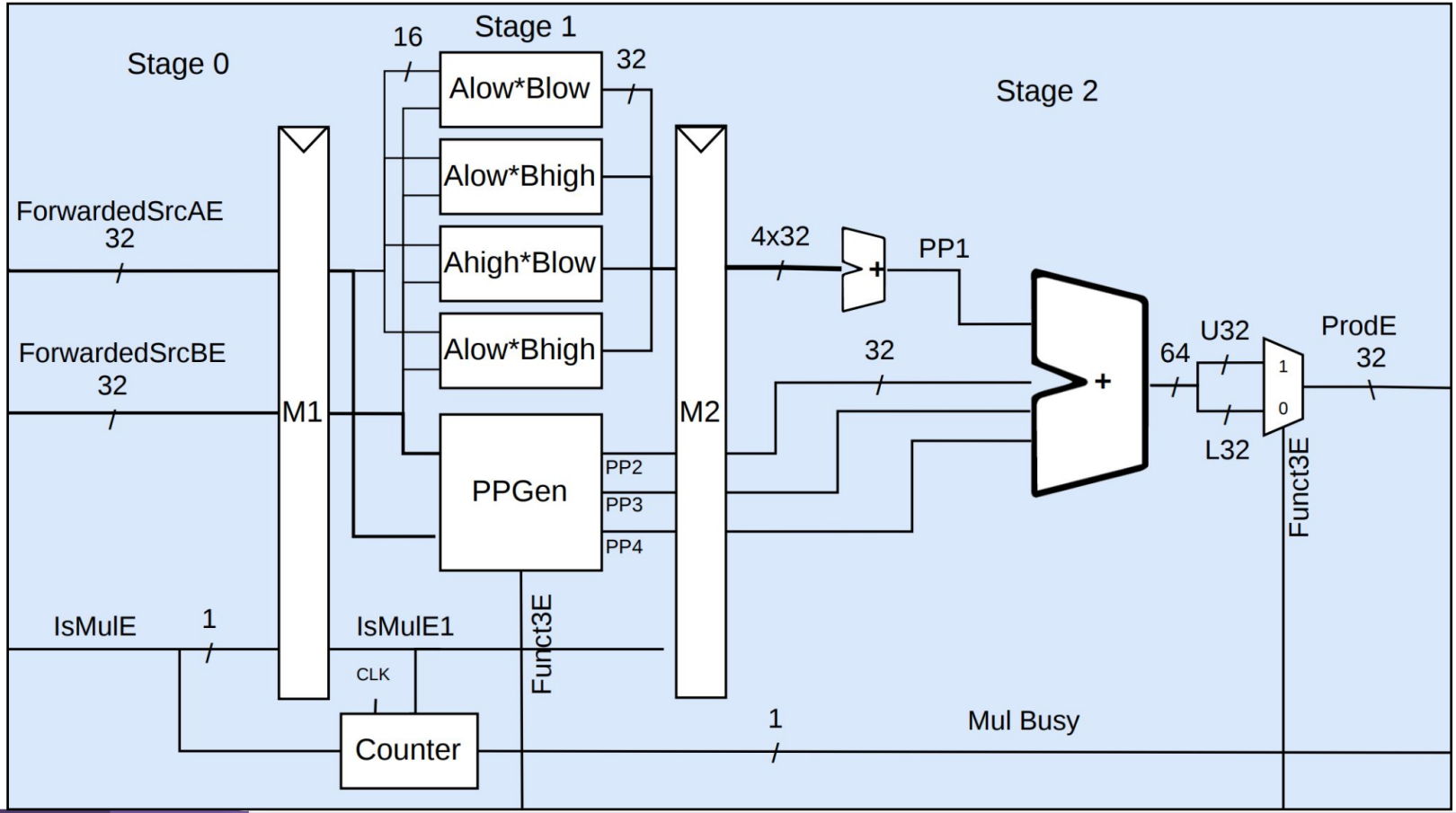
Signed * Signed   = P' + (-PA - PB)*2^(XLEN-1) + PP*2^(2XLEN-2)

Signed * Unsigned   = P' + (PA - PB)*2^(XLEN-1) - PP*2^(2XLEN-2)

Unsigned * Unsigned = P' + (PA + PB)*2^(XLEN-1) + PP*2^(2XLEN-2)

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Multiply Unit



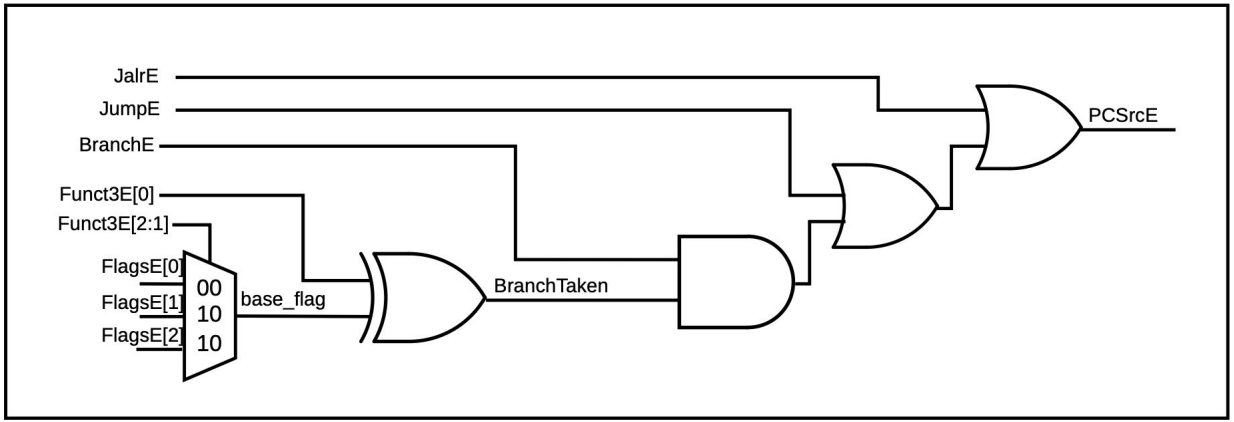
Branch Logic

Bit-level symmetry optimization

XOR gate to handle both “true” and “inverted” condition

Funct3[2:1]	Selected Base Flag	Instructions Covered
00	FlagsE[0] (Zero)	BEQ, BNE
10	FlagsE[1] (Less Than)	BlT, BGE
11	FlagsE[2] (Less Than Unsigned)	BLTU, BGEU

Branch Logic



Backwards Branch Predictor

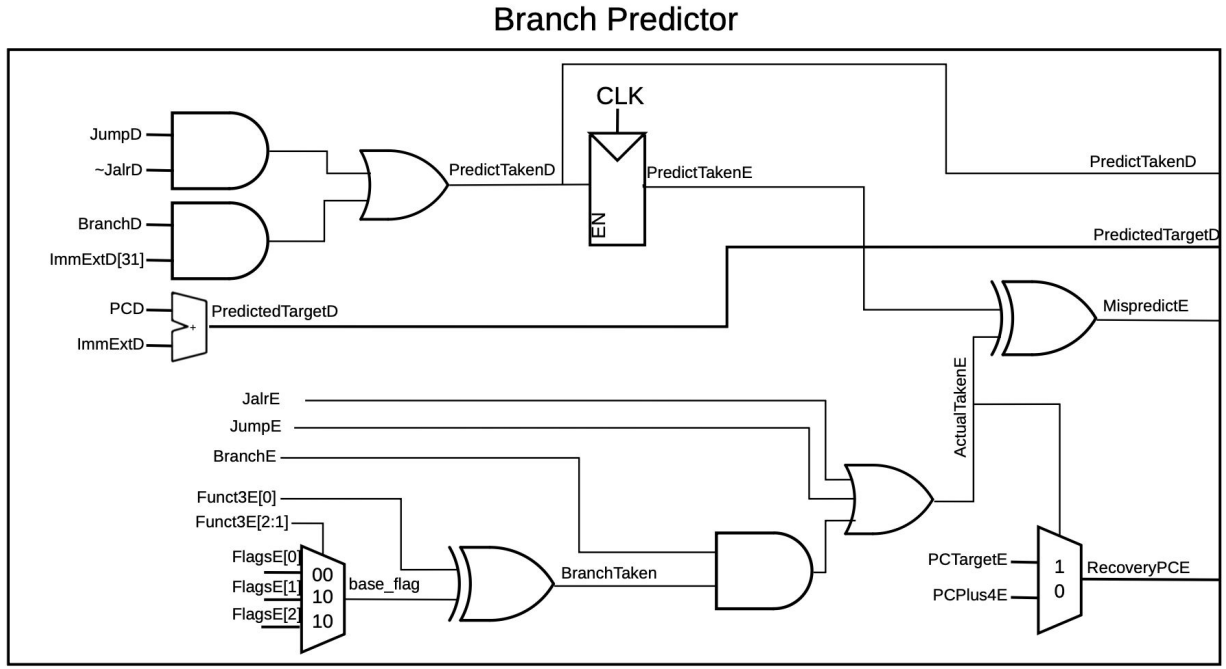
Builds off Branch Logic

Predict branch taken if the offset is negative (ImmExtD[31] = 1)

Original: 2 cycle for branch taken

Backwards taken predictor:
1 cycle for branch taken, 2 for not taken

Coremark/MHz 2.83 → 2.91



Synthesis

```
SYNTH_MAKE_FLAGS ?= \
   MAXOPT=1 \
   FREQ=333 \
```

Synth.tcl:

set_max_area 0

set_critical_range 0.10
$current_design

```
if { $maxopt == 1 } {
  ungroup -all -simple_names -flatten
  compile_ultra -gate_clock -retime
  optimize_registers
}
```

ungroup -all -simple_names -flatten:
Removes all module hierarchy and flattens the entire design into a single flat netlist before compilation. This gives DC maximum freedom to optimize across module boundarie

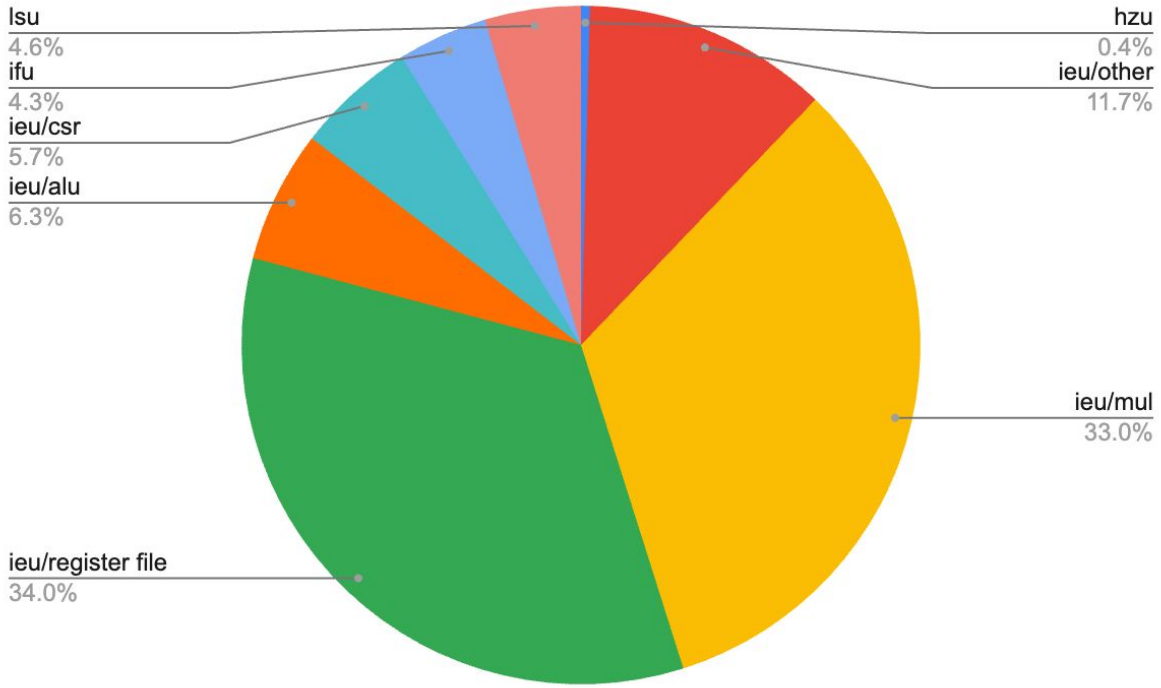
compile_ultra -retime:
Uses the full suite of optimization algorithms including datapath optimization, constant propagation, and logic restructuring. Allows the design compiler to move registers across combinational logic to balance path delays.

optimize_registers:
A second pass that runs after compile_ultra specifically focused on register placement. This includes moving registers to reduce setup time, merging redundant registers, and pipelining registers more efficiently across the flattened netlist.

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Area Minimization

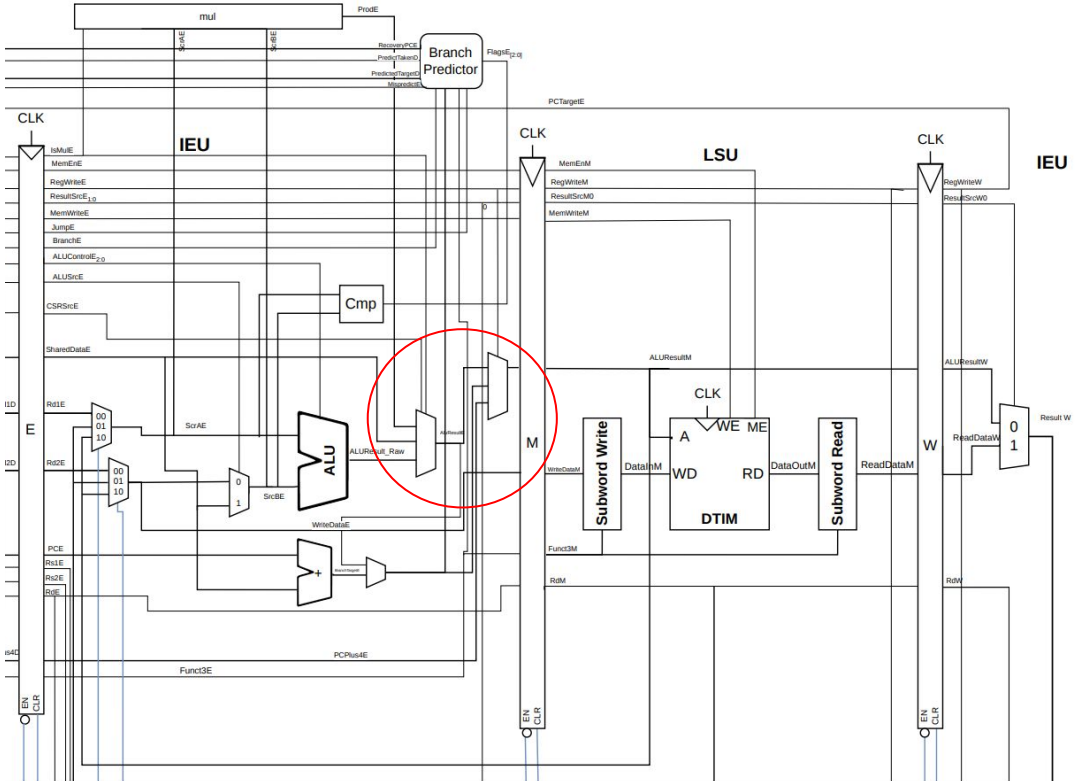
Total Area (μm^2): 188,099.94



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Launchpad
Max Conine (2 weeks ago)
Ln 89, Col 7
Tab Size: 4
UTF-8
LF
System Verilog

Area Minimization

CSR file reduced size
Compressed Signals onto the ALU result line

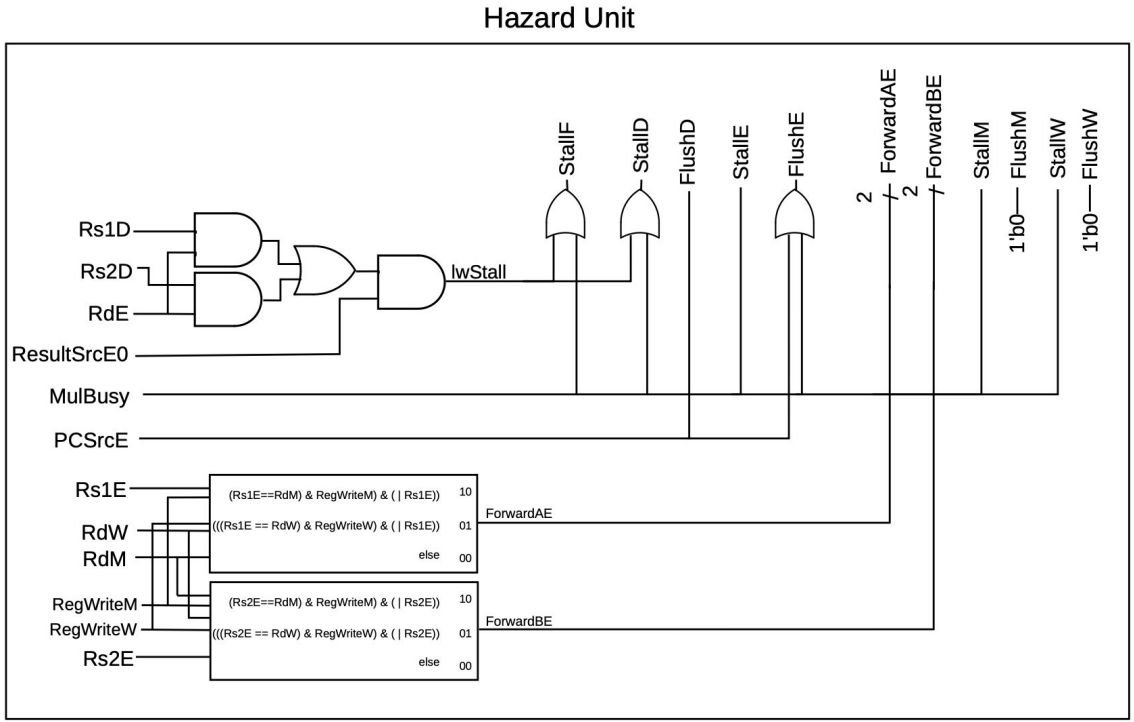


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Hazarding

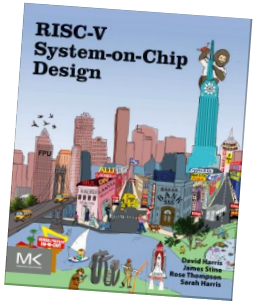
lwStall
MulBusy



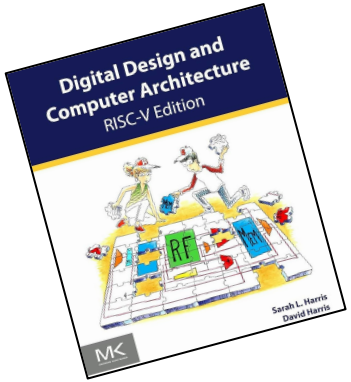
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Final Score

Gemini



4,814.14







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SOC_Final_P
Max Conine (2 weeks ago)

Progress Table

	Initial	Checkpoint 1	Checkpoint 2	Final
Figure of Merit	456.99	777.19 (+41.19%)	4,208.96 (+441.57%)	4814 .14 (+12.588%)
Crit Path Length	8.24 ns	5.101549 (-38.1%)	3.017745 (-40.98%)	3.017745 (No Change)
Area	221,069.87 μm^2	282,589.71 (+27.82%)	196,409.57 (-30.497%) μm^2	188,099.94 (-4.231%) μm^2
Coremark Score	3.79	3.07 (-18.997%)	2.83 (-7.818%)	2.91 (+2.827%)
CPI	1	1.140 (+14%)	1.24 (+8.77%)	1.190 (-4.032%)
Execution Time	27,494,271	16,574,458.26 (-39.71%)	10,655,014.82 (-35.714%)	10,336,280.59 (-2.991)

Final Score

Score: 4,814.14

Formula: 1e+30 / ((cycles x critical_path)^3 x area^1)

MTIME (cycles): 3,425,167

Critical Path Length (ns): 3.017745 [from timing.rep data arrival time]

Total Area (µm²): 188,099.94

Execution Time (ns): 10,336,280.59

COREMARK BENCHMARK DIAGNOSTICS

Status: PASSED

CoreMark Size: 666

Iterations: 10

Total Cycles (MTIME): 3,425,167

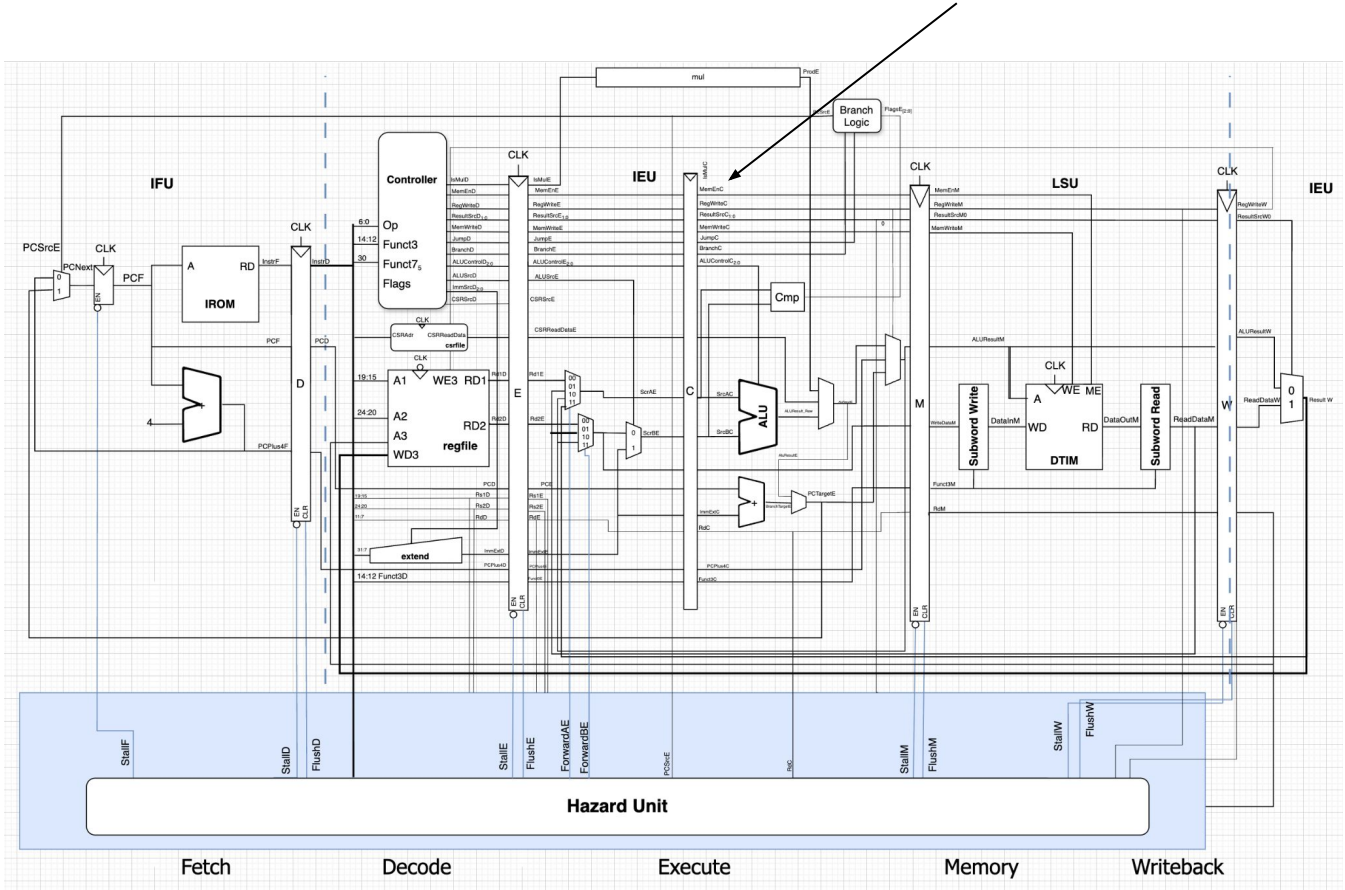
Total Instructions (MINSTRET): 2,864,107

CPI (Cycles Per Instruction): 1.190

COREMARK/MHz Score: 2.91

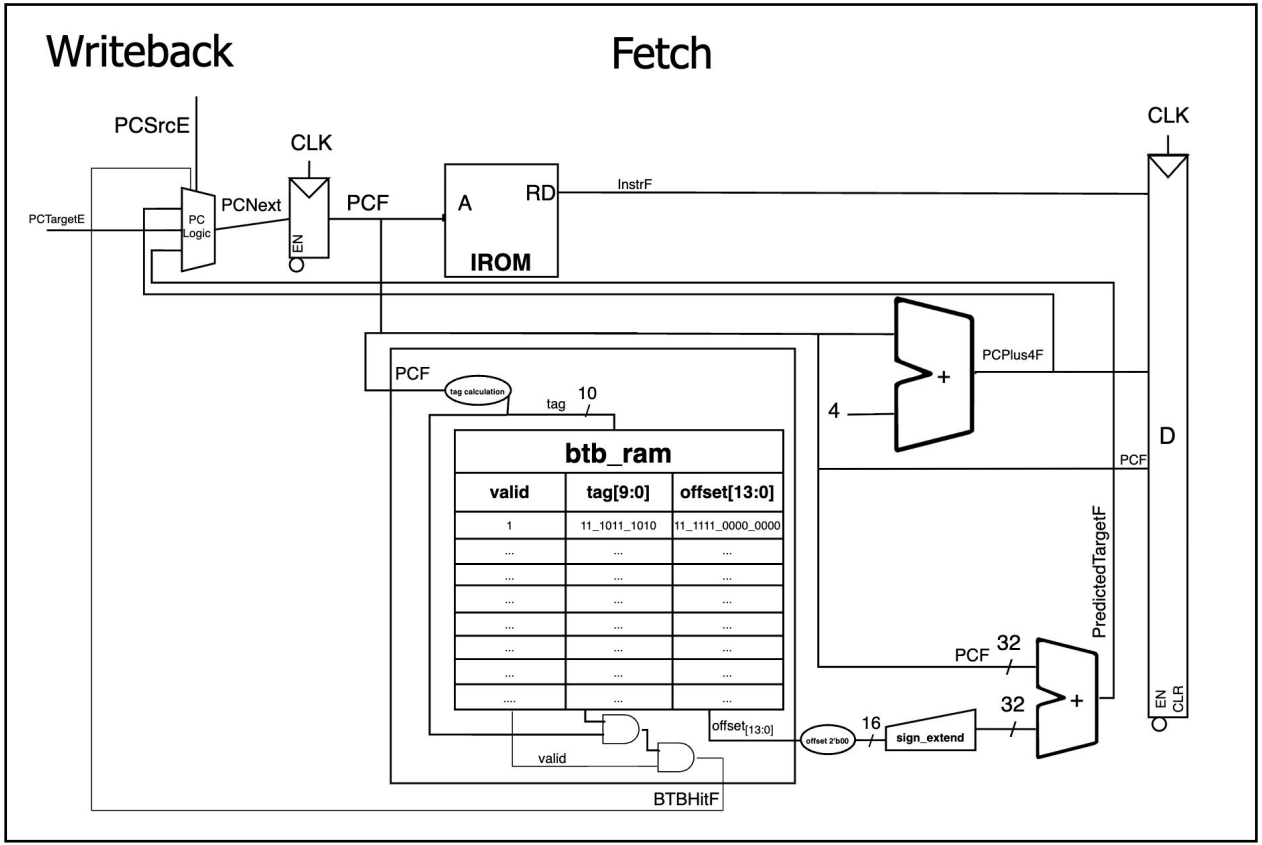
Future Work

6-stage processor



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0 0 0
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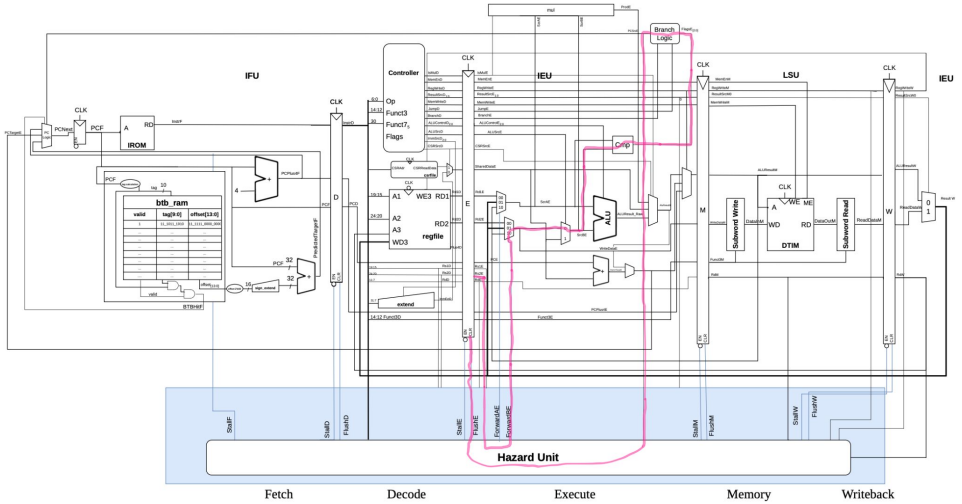
Branch Prediction Previous Design

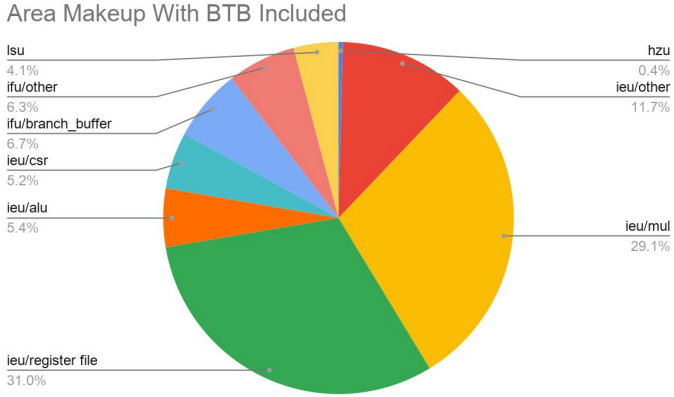


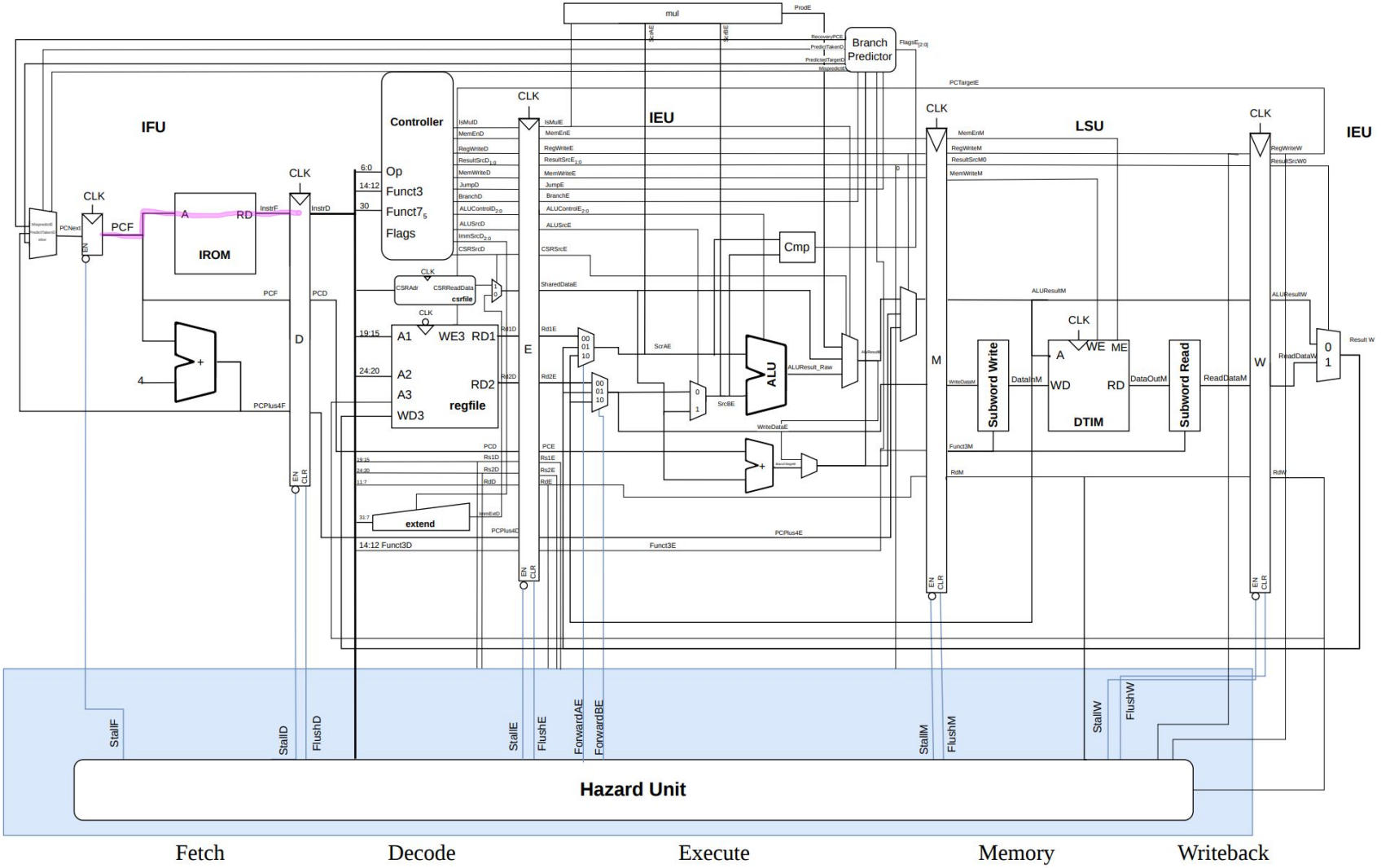
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Branch Prediction Findings

Number of index bits	Coremark / MHz score	FOM
2 (4 slots)	2.88	3,749
3 (8 slots)	2.95	4,085.3
4 (16 slots)	3	4,000
5 (32 slots)	3.03	3227






mul
Branch Predictor
IFU
IEU
LSU
IEU
Controller
Op
Funct3
Funct7
Flags
IROM
PCF
PCNext
InstrF
InstrD
regfile
extend
csrfile
ALU
Cmp
Subword Write
Subword Read
DTIM
Hazard Unit
StallF
StallD
FlushD
StallE
FlushE
ForwardAE
ForwardBE
StallM
FlushM
StallW
FlushW
Fetch
Decode
Execute
Memory
Writeback